

SKILL_WEEK - 07 (SESSION NUMBER - 13)

Task 1: To parse input redirection, open files, redirect stdin, handle missing files, restore streams, and test file inputs.

```
UW PICO 5.09

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>

int main()
{
    int saved_stdin;
    int fd;
    char word[100];

    saved_stdin = dup(STDIN_FILENO);

    fd = open("input.txt", O_RDONLY);

    if (fd < 0)
    {
        perror("Input file error");
        return 1;
    }

    dup2(fd, STDIN_FILENO);
    close(fd);

    printf("Reading from redirected input file:\n");

    while (scanf("%99s", word) == 1)
    {
        printf("%s\n", word);
    }

    dup2(saved_stdin, STDIN_FILENO);
    close(saved_stdin);

    printf("Standard input restored successfully.\n");

    return 0;
}
```

OUTPUT:

```
amrutha@Amruthas-MacBook-Air OSSP % echo "Operating Systems input redirection test" > input.txt
amrutha@Amruthas-MacBook-Air OSSP % gcc src/session13_task1.c -o bin/session13_task1
./bin/session13_task1
Reading from redirected input file:
Operating
Systems
input
redirection
test
Standard input restored successfully.
```

Task 2: To parse output redirection, create output files, redirect stdout, handle permissions, verify output data, and test redirection.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>

int main()
{
    int saved_stdout;
    int fd;

    saved_stdout = dup(STDOUT_FILENO);

    fd = open("output.txt",
              O_WRONLY | O_CREAT | O_TRUNC,
              0644);

    if (fd < 0)
    {
        perror("Output file error");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This line is redirected to output.txt\n");
    printf("Output redirection is working successfully.\n");

    fflush(stdout);

    dup2(saved_stdout, STDOUT_FILENO);
    close(saved_stdout);

    printf("Standard output restored successfully.\n");

    return 0;
}
```

OUTPUT:

```
amrutha@Amruthas-MacBook-Air OSSP % gcc src/session13_task2.c -o bin/session13_task2
amrutha@Amruthas-MacBook-Air OSSP % ./bin/session13_task2
Standard output restored successfully.
amrutha@Amruthas-MacBook-Air OSSP % cat output.txt
This line is redirected to output.txt
Output redirection is working successfully.
amrutha@Amruthas-MacBook-Air OSSP %
```

(SESSION NUMBER - 14)

Task 1: To parse the append operator, open files in append mode, redirect streams, preserve existing data, validate appends, and test scenarios.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>

int main()
{
    int saved_stdout;
    int fd;

    saved_stdout = dup(STDOUT_FILENO);

    fd = open("append_output.txt",
              O_WRONLY | O_CREAT | O_APPEND,
              0644);

    if (fd < 0)
    {
        perror("File open error");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("New appended line from Session 14 Task 1\n");
    fflush(stdout);

    dup2(saved_stdout, STDOUT_FILENO);
    close(saved_stdout);

    printf("Data appended successfully.\n");

    return 0;
}
```

OUTPUT:

```
amrutha@Amruthas-MacBook-Air OSSP % echo "Existing data in file" > append_output.txt
amrutha@Amruthas-MacBook-Air OSSP % gcc src/session14_task1.c -o bin/session14_task1
amrutha@Amruthas-MacBook-Air OSSP % ./bin/session14_task1
Data appended successfully.
amrutha@Amruthas-MacBook-Air OSSP % cat append_output.txt
Existing data in file
New appended line from Session 14 Task 1
amrutha@Amruthas-MacBook-Air OSSP %
```

Task 2: To parse stderr redirection, redirect error streams, capture error output, verify file contents, handle failures, and test error cases.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>

int main()
{
    int saved_stderr;
    int fd;

    saved_stderr = dup(STDERR_FILENO);

    fd = open("error_output.txt",
              O_WRONLY | O_CREAT | O_TRUNC,
              0644);

    if (fd < 0)
    {
        perror("File open error");
        return 1;
    }

    dup2(fd, STDERR_FILENO);
    close(fd);

    fprintf(stderr, "Error: sample redirected error message.\n");

    fflush(stderr);

    dup2(saved_stderr, STDERR_FILENO);
    close(saved_stderr);

    printf("Standard error restored successfully.\n");

    return 0;
}
```

OUTPUT:

```
[amrutha@Amruthas-MacBook-Air OSSP % gcc src/session14_task2.c -o bin/session14_task2
[amrutha@Amruthas-MacBook-Air OSSP % ./bin/session14_task2
Standard error restored successfully.
[amrutha@Amruthas-MacBook-Air OSSP % cat error_output.txt
Error: sample redirected error message.
amrutha@Amruthas-MacBook-Air OSSP %
```